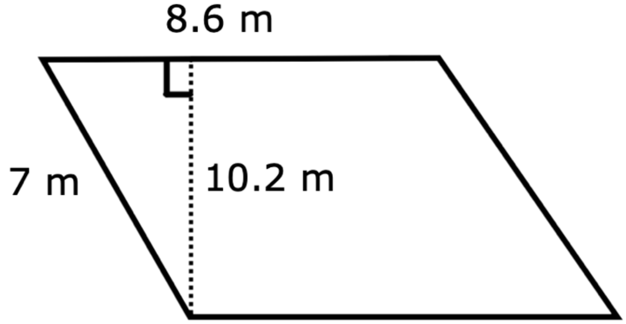
Grade 6 Accelerated
Unit 7
Lesson 4 Practice

Name____Answer Key_____
Date____________________
Period___________________

A parallelogram with dimensions labeled is shown below for questions 1-2.



1. Which dimension is not needed to find the area of the parallelogram? Explain.
   Area = base · height. The length of the slanted side is neither the base nor height, so it is not needed.

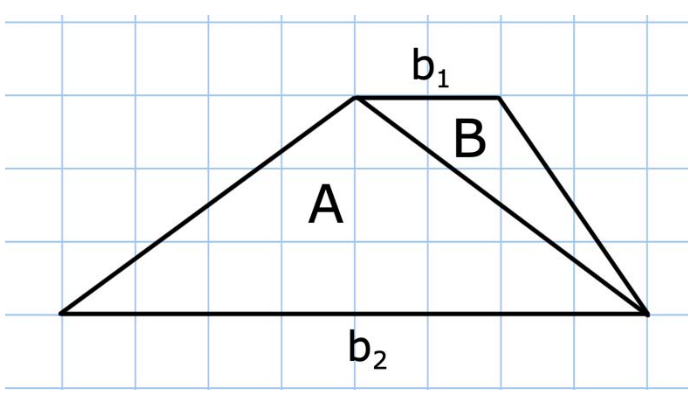
2. Fill in the blanks to find the area of the parallelogram.

   Area of parallelogram $= b \cdot h$

   Area = __8.6__ · __10.2__

   Area = __87.72__ m^2

A trapezoid can be decomposed into two triangles with the same height. Use the diagram below for questions 3-6.



3. The base of triangle A, b_2, is __8___ units and its corresponding height is ___3___ units. Therefore, the area of triangle A is ___12____.
   $\frac{1}{2} \cdot 8 \cdot 3 = 12$

4. The base of triangle B, b_1, is __2___ units and its corresponding height is ___3___ units. Therefore, the area or triangle B is ___3____.
   $\frac{1}{2} \cdot 2 \cdot 3 = 3$

5. The area of the trapezoid is ___15___ units^2.
$12 + 3 = 15$

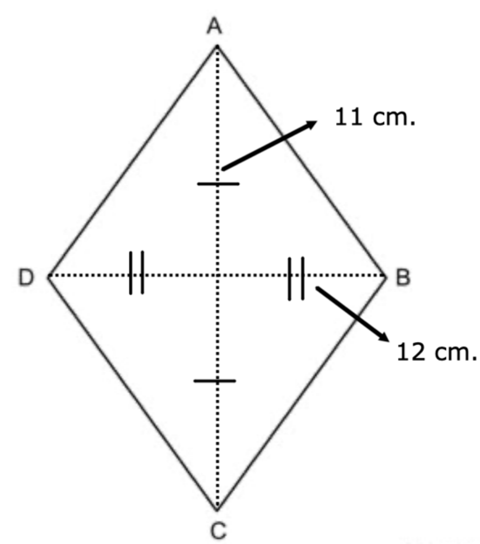
6. Use the formula for area of a trapezoid to verify your solution is correct.

Area = $\frac{1}{2}h(b_1 + b_2)$.

Area = $\frac{1}{2}$ __3__ (__2__ + __8__)

Area = __15__ units^2

Rhombus $ABCD$ is shown. Diagonal BD is 12 centimeters and diagonal AC is 11 centimeters. Use the diagram below for questions 7-10.



7. The base of triangle ABC is __11__ cm. and its corresponding height is __6__ units. Therefore, the area of triangle ABC is __33__.
$\frac{1}{2} \cdot 11 \cdot 6 = 33$

8. The base of triangle ADC is __11__ cm. and its corresponding height is __6__ units. Therefore, the area or triangle ADC is __33__.
$\frac{1}{2} \cdot 11 \cdot 6 = 33$

9. The area of the rhombus is ___66___ cm^2.
$33 + 33 = 66$

10. Use the formula for area of a rhombus to verify your solution is correct.

Area = $\frac{1}{2}(d_1)(d_2)$.

Area = $\frac{1}{2}$(__11__)(__12__)

Area = __66__ units^2